Emadeldeen Hamdan

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Education

University of Illinois Chicago — **Jan. 2023 – Present**
Ph.D., Electrical and Computer Engineering — *Chicago, IL*

- Researching efficient deep learning and computer vision methods for remote sensing and medical imaging applications, with a focus on lightweight AI architectures and foundation models.

University of Illinois Chicago — **Sep. 2020 – May 2022**
Bachelors, Electrical and Computer Engineering — *Chicago, IL*

- Magna Cum Laude Honors. Dean's List 2021, 2022.

Professional Summary

Ph.D. Candidate in Electrical and Computer Engineering at the University of Illinois Chicago developing efficient AI, computer vision, and signal processing methods for remote sensing and medical imaging. Research interests include lightweight deep learning architectures, foundation models, accelerated MRI reconstruction, prostate MRI segmentation, and prostate cancer analysis. Expected Ph.D. graduation: May 2027.

Experience

Springer Nature — **July 2025 – Present**
Assistant Editor — *Chicago, IL*

- Reviewed machine learning and signal processing manuscripts in collaboration with Editor-in-Chief Prof. Ahmet Enis Cetin, assessing technical quality, novelty, and clarity.

Northwestern University — **May 2025 – Present**
Research Assistant – Machine and Hybrid Intelligence Lab — *Chicago, IL*

- Develop novel deep learning methods for prostate MRI analysis, including accelerated MRI reconstruction, prostate zonal segmentation, lesion segmentation, and clinically significant prostate cancer (csPCa) detection.
- Design efficient CNN-, Transformer-, and Mamba-based architectures, leveraging transfer learning and foundation models to improve accuracy, robustness, and computational efficiency.
- Conduct research under Prof. Ulas Bagci and Prof. Ahmet Enis Cetin, contributing to publications in leading medical imaging and AI conferences.

University of Illinois Chicago — **Jan 2023 – Present**
Research & Teaching Assistant, Electrical and Computer Engineering — *Chicago, IL*

- **Research Assistant**
  - Developed efficient deep learning and signal processing algorithms for computer vision, remote sensing, and real-time biomedical signal analysis.
  - Developed AI models for wildfire and peatland fire detection from visible, infrared, and multispectral imagery, including large-scale dataset development and cross-dataset evaluation.
  - Developed a real-time deep learning framework for instantaneous neural phase estimation from electrophysiological recordings, resulting in an accepted IEEE Transactions on Biomedical Engineering (TBME) publication.
- **Teaching Assistant**
  - Supported undergraduate and graduate courses in Digital Signal Processing and Neural Networks through office hours, laboratory instruction, and assignment development.

Handshake AI — **Jan 2026 – May 2026**
AI Trainer – Generative AI Model Evaluation — *Chicago, IL*

- Evaluated large-scale generative AI model outputs across multimodal tasks, improving response quality and factual accuracy.

L3DFX LLC — **May 2022 – Jan 2023**
Electrical Engineer — *Bolingbrook, IL*

- Designed and tested signal processing circuits and embedded systems for real-world applications.
- Integrated microcontroller-based solutions for control systems, improving system reliability and performance.

Selected Publications

Zonal-Aware Prostate Cancer Detection via Hadamard-Fused Mamba-UNet.
Introduced a transform-enhanced state-space architecture combining Hadamard representations with Mamba-based models, achieving real-time prostate zonal segmentation (∼0.01s) and improved cancer detection performance. Submitted to *MICCAI Workshops*, 2026.

Large-Scale Open-Source Image and Video Dataset for Robust Wildfire Detection and Classification.
Introduced a large-scale multimodal wildfire benchmark dataset and proposed an efficient Hadamard transform-based feature fusion mechanism for robust wildfire detection. Accepted to *ICIP HydroImaging Workshop*, 2026.

Hybrid Multi-Dimensional MRI Prostate Cancer Detection via Hadamard-Based Bias Correction and Residual Networks.
Designed a hybrid transform-based pipeline for MRI bias correction and classification, improving cancer detection accuracy by 30%. Accapted to *EMBC*, 2026.

Discrete Cosine Transform Based Decorrelated Attention for Vision Transformers.
We introduce two methods leveraging the Discrete Cosine Transform (DCT) to enhance the efficiency and performance of Vision Transformers. Accapted to *IJCAI*, 2026.

Real-time Instantaneous Phase Estimation Using a Deep Dual-Branch Complex Neural Network.
Proposed a lightweight neural architecture for real-time instantaneous phase estimation of neural oscillations, with FPGA implementation for low-latency deployment. Published in *IEEE Transactions on Biomedical Engineering (TBME)*, 2025.

Reconfigurable Electromagnetically Unclonable Functions Based on Graphene RF Modulators.
Assisted in the development of a robust hardware security primitive leveraging electromagnetic PUFs for device authentication and cryptographic identification. Published in *ACS Nano*, 2025.

Wildfire Detection via Transfer Learning: A Survey.
Surveyed transfer learning approaches for large-scale wildfire detection, highlighting challenges in generalization and dataset bias. Published in *Signal, Image and Video Processing*, 2024.

Technical Skills

Programming: Python, C, MATLAB
Machine Learning: PyTorch, TensorFlow, State-Space Models (Mamba), Transform-based Learning (Hadamard, DCT), Computer Vision, Representation Learning, Generative Models (LLMs, VLMs)
Systems & Efficiency: GPU Training (CUDA), Hardware-Aware ML, Low-Complexity Model Design, FPGA Deployment, High-Level Synthesis (HLS)
Tools: Linux (CLI), Git, VS Code

Awards and Leaderships

IEEE Communication Society — **May 2025 – Present**
Secretary – UIC Student Branch Chapter — *Chicago, IL*

- Organizing technical events and workshops focused on AI, signal processing, and communications systems.

NASA Beyond the Algorithm Challenge — **Sep 2025**

- **Top 9 Finalist** — Developed transform-based ML layers (Hadamard, Haar wavelets) for efficient computing in quantum and in-memory systems (Team NVSense).

NASA MSI Incubator Wildfire Climate Tech Challenge — **Mar 2024**

- **Runner-Up** — Built a computer vision system for wildfire situational awareness and early detection (Team FIRESENSE).

NSF I-Corps (POSE Program) — **Feb 2024**

- **Entrepreneurial Lead** — Led commercialization efforts for hardware-aware ML solutions (Team HLS4ML).